

PHISHSENSE — INTELLIGENT PHISHING LINK DETECTION SYSTEM

DECLARATION

I declare that this project, titled **PhishSense — Intelligent Phishing Link Detection System**, was developed as an academic project to investigate and implement techniques for detecting potentially malicious URLs. The system integrates machine learning, rule-based analysis and threat intelligence into a single application.

DEDICATION

This project is dedicated to my family, lecturers, colleagues and everyone who has supported me throughout my studies.

ACKNOWLEDGEMENT

I am grateful to my lecturer and the Department of Computer Science, KNUST, for their guidance and support throughout the development of this project. I also appreciate the support of my colleagues, friends and family.

ABSTRACT

Phishing is a major cybersecurity threat in which attackers use deceptive links to trick users into revealing sensitive information or visiting malicious websites. PhishSense is a web-based phishing URL detection system developed to help users assess the risk of a URL before interacting with it.

The system uses three main detection layers: **heuristic analysis, threat intelligence and machine learning**. The heuristic layer checks suspicious URL structures and patterns, the threat intelligence layer compares URLs against security databases, and the machine learning component analyses URL features to identify phishing patterns. The results from these layers are combined into an explainable risk score from 0 to 100.

The system also provides a user-friendly interface that presents a simple security verdict while allowing users to view advanced technical details.

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CHAPTER ONE: INTRODUCTION

1.1 Background of the Project

Phishing is a form of cyberattack where attackers create deceptive websites or URLs to trick users into providing information such as passwords, banking details and account credentials.

Many existing phishing detection approaches depend mainly on blacklists, predefined rules or machine learning. However, each approach has limitations. A blacklist may not immediately contain a newly created phishing URL, while rules may fail to detect unfamiliar patterns. Machine learning can identify patterns that are difficult to manually define but may also produce incorrect predictions.

PhishSense was therefore developed as a multi-layer URL analysis system that combines these approaches.

1.2 Problem Statement

Users may receive suspicious links through emails, social media, messaging applications and other online platforms. It can be difficult for non-technical users to determine whether a URL is safe simply by looking at it.

A system that combines multiple sources of evidence and presents the result in an understandable way can help users make safer decisions.

1.3 Aim of the Project

The aim of PhishSense is to develop a web-based system that analyses URLs and provides users with an understandable assessment of their potential phishing risk.

1.4 Novelty of the Project

The main contribution of PhishSense is its **multi-layer and explainable detection approach**.

Instead of depending on only one technique, PhishSense combines:

- **Heuristic analysis** — checks suspicious URL characteristics and patterns.
- **Threat intelligence** — checks external security databases for known threats.
- **Machine learning** — identifies phishing-related patterns in URL features.

- **Composite risk scoring** — combines the results into one score.
- **Explainable results** — shows users the reasons contributing to the result.

The project therefore focuses not only on detection, but also on **combining different detection approaches and making their results understandable to users**.

1.5 Specific Objectives

The objectives are to:

1. Develop a web interface for submitting URLs for analysis.
2. Implement heuristic URL analysis.
3. Integrate external threat intelligence sources.
4. Integrate a machine learning phishing classification model.
5. Develop a composite risk-scoring mechanism.
6. Provide understandable explanations for detection results.
7. Protect the analysis process using SSRF-related security controls.

1.6 Scope

The system focuses primarily on **URL-based phishing detection**. It analyses URL structure, suspicious keywords, domain characteristics, impersonation patterns, threat intelligence results and machine learning predictions.

The current system does not perform complete webpage content analysis or guarantee that every URL is safe.

1.7 Limitations

The system has several limitations:

- The machine learning model is trained on a relatively limited dataset.
- Threat intelligence results depend on the availability of external services.
- URL-based analysis alone cannot identify every phishing website.
- Attack techniques can change over time, requiring future model improvements.

1.8 Beneficiaries

The main beneficiaries are:

- Internet users who want to check suspicious links.
- Students and researchers studying cybersecurity.
- Organisations interested in basic URL security analysis.
- Security awareness and education programmes.

CHAPTER TWO: REVIEW OF RELATED SYSTEMS

2.1 VirusTotal

VirusTotal is an online security service that analyses files, domains and URLs using multiple security vendors.

Good features:

It provides results from many security vendors and offers broad threat intelligence coverage.

Bad features:

A URL may not yet be present in vendor databases, and the large amount of technical information may be difficult for ordinary users to understand.

2.2 Google Safe Browsing

Google Safe Browsing helps identify websites associated with malware, phishing and other unsafe activities.

Good features:

It has extensive infrastructure and can identify many known dangerous websites.

Bad features:

It primarily relies on Google's detection infrastructure and does not provide the same level of user-visible multi-layer explanation as PhishSense.

2.3 URLScan.io

URLScan.io provides detailed information about websites and their network activity.

Good features:

It provides detailed website and network analysis.

Bad features:

Its technical nature can make the results difficult for non-technical users to interpret.

2.4 OpenPhish

OpenPhish provides phishing intelligence and maintains feeds of identified phishing websites.

Good features:

It is useful for identifying known phishing URLs.

Bad features:

New phishing URLs may not immediately appear in the feed.

2.5 Comparison with PhishSense

The reviewed systems provide strong security capabilities, but PhishSense focuses on combining **heuristics, threat intelligence and machine learning** into one scoring pipeline while presenting the outcome in a simpler form.

The system is therefore not intended to replace established security services. Instead, it demonstrates how multiple detection techniques can be integrated into a single explainable application.

CHAPTER THREE: METHODOLOGY

3.1 Architecture of the Proposed System

PhishSense follows a modular architecture consisting of:

User → React Frontend → Node.js Backend → Detection Layers → Risk Scoring → Results

The three main detection layers are:

1. Heuristic Rule Engine
2. Threat Intelligence
3. Machine Learning Model

Their results are combined by the backend and returned to the frontend.

3.2 Functional Requirements

The system should:

- Accept a URL from the user.
- Validate the submitted URL.
- Analyse URL characteristics.
- Query available threat intelligence services.

- Perform machine learning analysis.
- Calculate a risk score.
- Display the final classification.
- Display reasons contributing to the result.
- Provide advanced technical information.

3.3 Non-Functional Requirements

The system should be:

- Usable by non-technical users.
- Responsive across different screen sizes.
- Secure against unsafe URL requests.
- Modular and maintainable.
- Reliable when individual detection services are unavailable.

3.4 Project Method

An iterative development approach was used. The system was developed in separate components and progressively integrated and tested.

3.5 Machine Learning Component

The machine learning component uses a **Random Forest classifier** for URL classification.

Rather than developing a new neural network architecture from scratch, the project integrates an existing machine learning approach into the PhishSense detection pipeline. The work involved preparing URL features, training/evaluating the classifier, saving the trained model and connecting it to the Node.js application for real-time prediction.

The model analyses URL characteristics such as:

- URL length
- hostname structure
- number of subdomains
- suspicious keywords
- HTTPS usage

- IP-based hosts
- suspicious TLDs
- URL shorteners
- brand similarity
- homoglyph indicators

The model produces a phishing probability which is converted into a maximum of **30 AI risk points**.

3.6 Risk Scoring

PhishSense combines the three detection layers:

Final Risk Score = Heuristic Score + Threat Intelligence Score + AI Score

The result is constrained between 0 and 100.

The scoring design allows one layer to compensate for limitations in another. For example, a newly created phishing URL that is not yet present in threat intelligence feeds may still receive a high score because of suspicious URL patterns detected by the heuristic and ML layers.

CHAPTER FOUR: IMPLEMENTATION, TESTING AND RESULTS

4.1 Technologies Used

Frontend

- React
- Vite
- JavaScript
- Tailwind CSS

Backend

- Node.js
- Express.js

Machine Learning

- Python

- Random Forest
- Custom URL feature extraction

Threat Intelligence

- VirusTotal
- OpenPhish

4.2 System Modules

The major modules are:

1. **URL Input Module** — accepts and validates URLs.
2. **Heuristic Analysis Module** — evaluates predefined suspicious URL patterns.
3. **Threat Intelligence Module** — obtains reputation information from external sources.
4. **AI/ML Module** — predicts phishing probability from URL features.
5. **Risk Scoring Module** — combines the detection results.
6. **Results Module** — presents the final risk level and explanations.
7. **Security Module** — protects URL fetching operations against unsafe destinations.

4.3 Testing

The system was tested using legitimate URLs and URLs containing suspicious characteristics such as:

- Typosquatting
- Suspicious keywords
- IP-based hosts
- Suspicious TLDs
- Multiple subdomains
- URL shorteners
- Brand impersonation

The testing demonstrated that the different detection layers can identify different types of suspicious URL characteristics.

4.4 Results

Example results included:

URL Type	Result
Legitimate websites	Low Risk
Typosquatting URLs	Suspicious
IP-based phishing URLs	High Risk
URLs with multiple suspicious keywords	High Risk
URLs combining impersonation and suspicious structures	High Risk

The machine learning component also demonstrated strong performance on its test dataset. However, these results should be interpreted as **benchmark results rather than a guarantee of real-world detection accuracy**, since phishing techniques continuously change.

CHAPTER FIVE: FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

5.1 Findings

The project found that combining different detection approaches provides more useful information than relying on a single method.

Heuristics are useful for identifying known suspicious structures, threat intelligence is useful for identifying known malicious URLs, and machine learning can identify patterns from previously observed examples.

The combination also improves explainability because users can see which layer contributed to the final risk score.

5.2 Conclusion

PhishSense successfully demonstrates a multi-layer approach to phishing URL detection. The system combines heuristic analysis, threat intelligence and machine learning within a single web application and presents the result through an understandable risk score.

The project also demonstrates that cybersecurity systems can provide both automated detection and explanations that help users understand why a URL was considered suspicious.

5.3 Challenges

Some challenges encountered during development included:

- Integrating different technologies and services.
- Handling external threat intelligence responses.
- Connecting the Python ML component to the Node.js backend.
- Protecting the system from unsafe URL destinations.
- Designing technical security results in a way that remains understandable to ordinary users.

5.4 Lessons Learnt

The project provided practical experience in:

- Web application development.
- Cybersecurity and phishing detection.
- Machine learning integration.
- API integration.
- Backend and frontend communication.
- Secure URL processing.
- System testing and documentation.

5.5 Recommendations for Future Work

Future versions could:

1. Increase the size and diversity of the training dataset.
2. Retrain the model regularly using newer phishing data.
3. Add webpage HTML and visual analysis.
4. Add more threat intelligence sources.
5. Improve detection of newly emerging phishing techniques.
6. Develop browser-extension support for real-time protection.

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